

✓
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 doubts
 exercise
 32 bits.
 int
 1 → 001
 0 → 000
 [-1, 0]
 xor = -1
 -1 → flip → 110
 + 1 → 001
 110 + 001 = 111
 { 111 }
 1 → 001
 xor
 1 1 1
 1 0 0 0
 1 1 1
 MSB
 -ve no.
 -ve
 (1 1 1) → 000
 + 1
 000 + 1 = 001
 xor = - (001)

$\overset{2}{0} \dots \overset{K}{n-1}$ } $\rightarrow$ linear search
 $K \rightarrow i \rightarrow \text{index}$

$\overset{2}{0} \dots \overset{K}{n-1}$

$K \rightarrow \oplus$
 $\textcircled{1}$

searching?

$K = 6$
how?

search

find

searching

$\textcircled{I}$ $\rightarrow$ $[I_1, I_2, I_3, I_4]$

$\textcircled{4} \rightarrow \text{ans}$

$[\underset{0}{1} \underset{1}{5} \underset{2}{2} \underset{3}{4} \underset{4}{6} 3 5]$

Time $\rightarrow$ WC
 complexity

$\overset{2}{0} \dots \overset{K}{n-1}$
 $\textcircled{n}$

$\textcircled{n}$ $K \oplus$
 $|O(n)|$



if(flag == false)
print(-1);

$K=5$ $n=6$
 $arr = [1, 2, 4, 9, 6, 5]$

```
flag = false;  
for (i = 0; i < n; i++) {  
    if (arr[i] == K) {  
        print(i);  
        flag = true;  
        break;  
    }  
}
```

$\approx O(n)$

$O(n^3)$

$n=1000$

1 - 2 5 - 3 2 4 → [1 5 2 3 9]

added

Kadane's
algo

max subarray sum
?

gen all subarrays

curr → sum

final ← (ans)

st
 $i=0 \dots n-1$
 $j=i \dots n-1$
 $k=i \dots j$

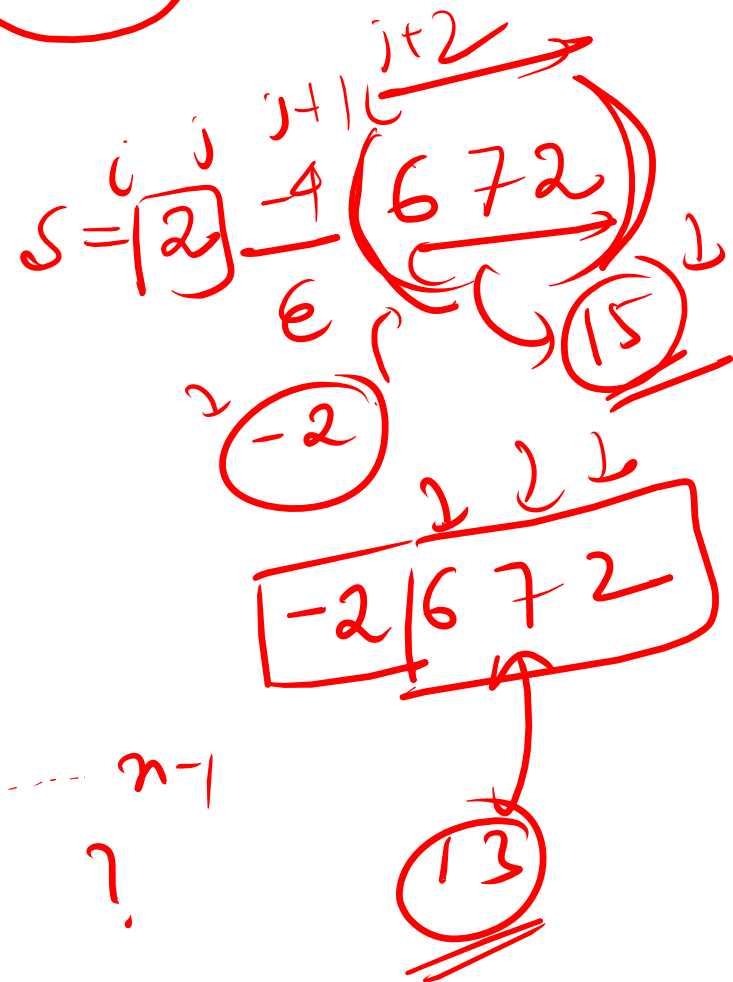
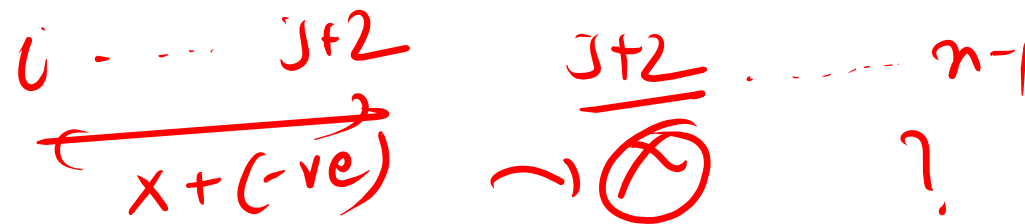
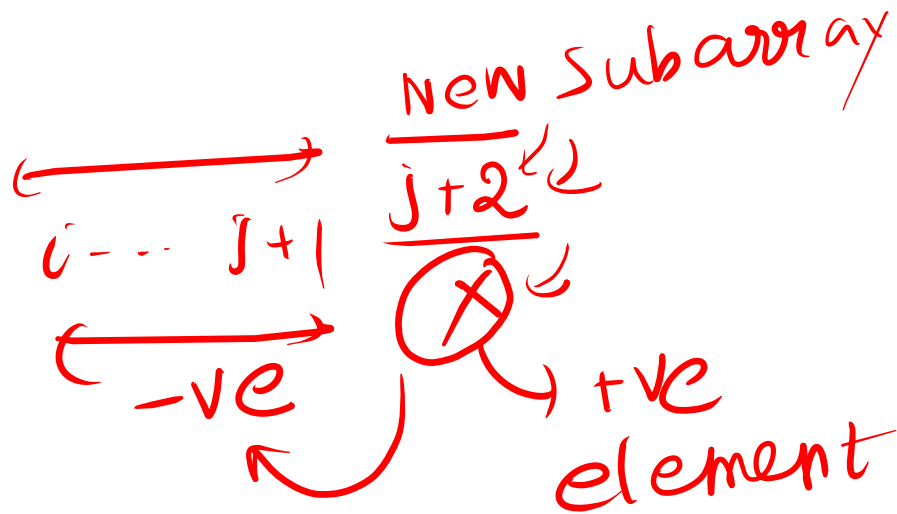
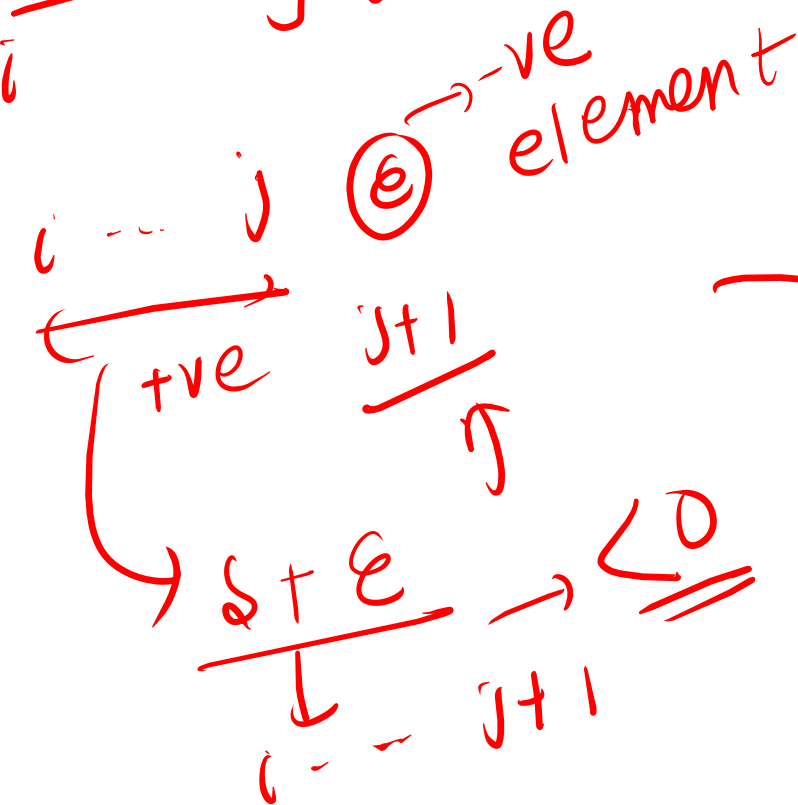
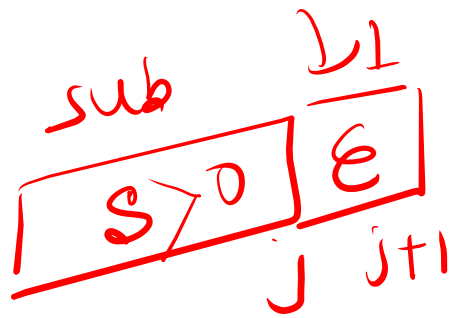
sum += arr[k]

ans = max(ans, sum);

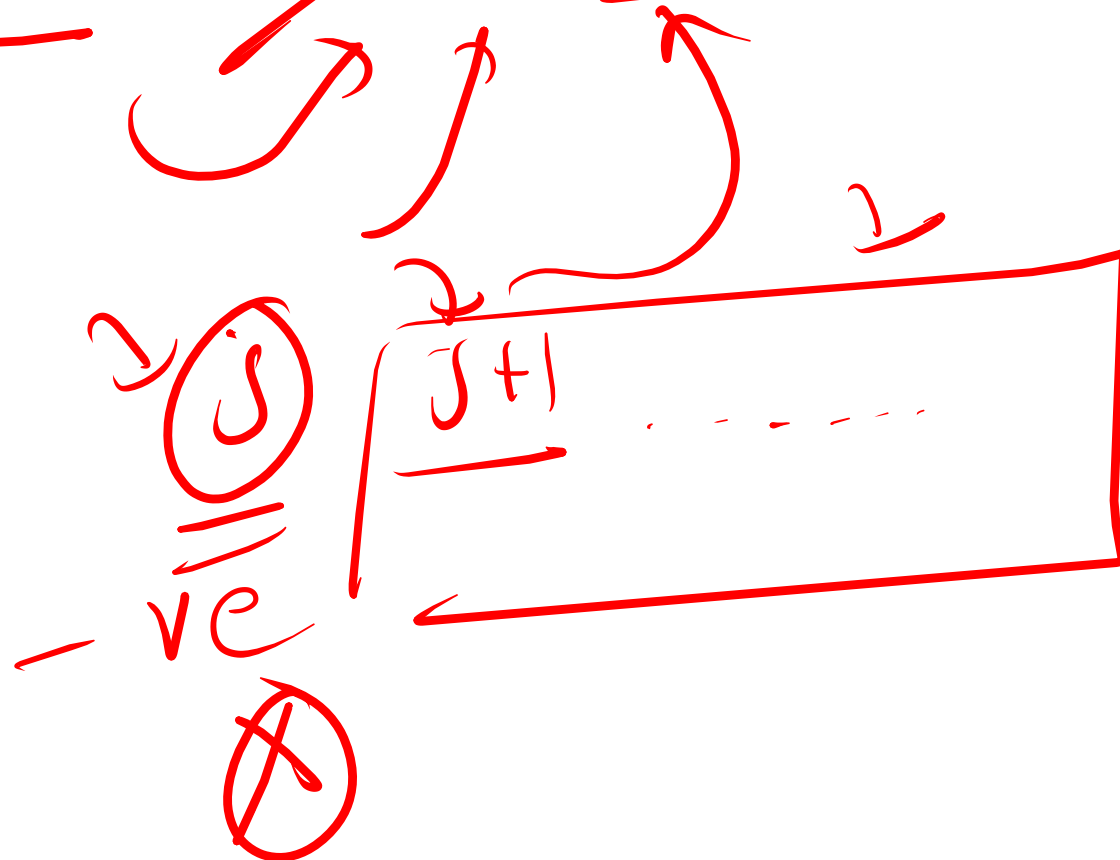
ans = final

1 - 2 5 - 3 2 4

15 $\rightarrow$ max / 13
 (+ve) + (-ve)



curr $\rightarrow$ ~~-ve~~ 0



$curr = \emptyset$ 1
 $ans =$
~~(INT_MIN)~~
~~1~~ ~~8~~
 $1 - 2 \begin{pmatrix} 5 & -3 & 2 & 4 \end{pmatrix} \oplus$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 \end{pmatrix} \rightarrow$ once $O(n) \rightarrow TC$
8

$\begin{matrix} & & j & j+1 & j+2 \\ & & \downarrow & & \downarrow \\ i & \boxed{s} & \boxed{e} & \boxed{x} & \end{matrix} \rightarrow x > 0$
 $st e < 0$
 $-ve$

$curr = 1 - 2$
 $() = \cancel{1} \cancel{\emptyset} \begin{matrix} 2 \\ 5 - 3 \end{matrix}$
 $= \underline{2} + 2$
 $= 4 + 4 = \underline{8}$

$s = \underline{0} + \cancel{8}$
ans